Lecture 5a
Data Warehouse Case Study

Data Warehousing and Data Mining
2026/2027 Semester 1

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Warehouse Design
Step 1. Select the Process
Step 2. Declare the Grain
Step 3. Choose the Dimensions
Step 4. Identify the Facts.
Case Study

A Sample Exercise to Complement Lecture Notes

- Supermarket Case Study
- Actions involved in steps 1-2
- Decisions at key points
- Promotion: a 4th dimension!



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Four-Step Dimensional Design Process

1. **Select the Process**: What process do you wish to measure?
2. Declare the grain of the business process
3. Choose the dimensions that apply to each fact table row
4. Identify the numeric facts that will populate each fact table row.



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Step 1. Identify/Select the Process

- A **process** is a natural activity captured by source data.
- Example business processes:
  How many Female students taking CA707, who live in Munster, are scoring firsts?
  How many frozen products, imported from France, are sold in January in April?
  How many patients, in Connaught, tested positive (for a number of ailments) in 2020? (these are dimensional queries!)



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Consistent Information

- Build a *single* dimensional model for every process in the organisation.
- Focus on processes/activities (not departments) to deliver *consistent* information more efficiently throughout the organisation.
- *Consistent*: this important property is studied in Fact & Dimension Conformance.



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Step 2. Declare the Grain

- Specify *exactly* what an individual fact table row represents.
- The **grain** conveys the level of detail associated with the fact table measurements.
- It provides the answer to the question: *How do you describe a single row in the fact table?*
  For our 3 examples, **grain is**:
  Every mark for every module for every student
  Every item sold (units), per country of origin, by month, by sales location
  Every positive test, for each disease, for a particular year in a specified region.



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Sample grain declarations

- An individual line item on a customer's retail sales ticket as measured by a scanner device
- A line item on a bill received from a doctor
- An individual boarding pass to get on a flight
- A daily snapshot of the inventory levels for each product in a warehouse
- A monthly snapshot for each bank account



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Step 3. Choose the Dimensions (for every measure)

- Dimensions emerge from asking: *How do analysts describe the data that results from the business process? At what level do scientists discuss the output from test sites? How are decisions made in hospitals when allocating . . . ?*
- Enrich fact tables with a robust set of dimensions representing all possible descriptions.
- Once the grain is known, dimensions can be identified easily.
- For each dimension, list all discrete, text-like attributes to populate each dimension table.
- *What dimensions were used in our 3 examples?*



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Step 4. Identify the numeric facts that will populate each fact table row.

- Facts are determined by answering the question: *What are we measuring?*
- All candidate facts in a design must be true to the grain defined in step 2.
- Facts that clearly belong to a different grain *must be in a separate fact table*.
- Typical facts are numeric additive figures such as items sold, tests conducted, students who participated, amount spent . . .



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Summary

Consider both business users' requirements *and* realities of source data, to make decisions regarding the 4 steps.

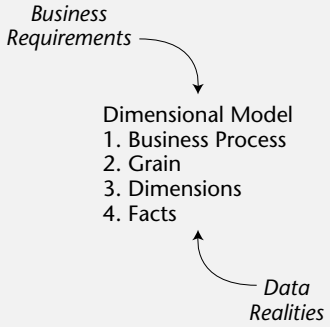


Figure 1: Key input to the four-step dimensional design process



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Case Study: Large Grocery Chain

- Business has 100 stores across multiple regions **(Dimension: Geo)**
- Each store has departments: grocery, frozen foods, dairy, meat, bakery and health/beauty aids.
- Each store has roughly 60,000 individual products **(Dimension: Product)**
- Bar codes are universal product codes (UPCs).



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Step 1. Select the Process

- The first dimensional model built should have **maximum impact**: it should answer the most pressing questions for the organisation.
- Assume management wants to *better understand* customer purchases as captured by the POS system: POS retail sales is selected.
- This data allows them to analyse *what products* are selling, in *which stores*, on *what days*, under what promotional *conditions*. **(Dimensions: Time, Promotion)**



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Step 2. Declare the Grain



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- *What level of data detail should be made available in the dimensional model?*
- Record data at its lowest, **most atomic grain** to capture detail.
- Non-atomic grain would be: weeks, not days; category not product; county not city; etc.

Atomic Data: Advantages



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1. Atomic data is **highly dimensional**.
2. The more detailed and atomic the fact measurement, the more things we know for sure (correct, certain).
3. Atomic data provides maximum analytic flexibility because it can be *constrained* and *rolled up* in every way possible.

Higher Grains: Disadvantages



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- Declare higher-level grains for a business process that represent an aggregation of the most atomic data.
- However, by selecting a higher-level grain, users are limited to fewer and/or potentially less detailed dimensions.
- The less granular model is immediately vulnerable to unexpected user requests to drill down into the details.
- It's easy to do a Rollup (Group-By) but you cannot do the reverse!

Step 3. Choose the Dimensions



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- Once the grain is chosen, the *date*, *product*, and *store* dimensions become obvious.
- We assume that the calendar date is the date value delivered to us by the POS system.
- Within the framework of the primary dimensions, we consider new dimensions such as the *promotion* under which the product is sold.
- Exam Questions usually look for a 4th dimension!



- Assume we decide on the following descriptive dimensions: `date`, `product`, `store`, and `promotion`.
- Include the POS transaction ticket number as a special dimension.
- A preliminary schema is shown in figure 2.
- *Do not begin* populating the dimension tables with descriptive attributes: instead complete the final step of the process (keep it simple for now!).

Preliminary Schema



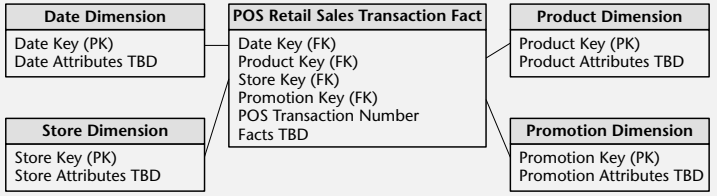


Figure 2: Preliminary retail sales schema

- What do we have at this point?
- Outline structure with connection points between tables.
- No details on dimensions (structure) or facts (what we'll measure).

Step 4. Identify the Facts



- There is a firm rule for this step: **the facts must be true to the grain**.
- Specifically, this is the individual line item on the POS transaction.
- The facts collected by the POS system include the sales quantity, per unit sales price, and the sales amount.
- The sales amount is: *quantity by unit price*.



- The current fact table is shown in Figure 3.
- Three of the facts, *sales quantity*, *sales amount*, and *cost amount*, are perfectly additive across all the dimensions.
- We can **slice** and **dice** the fact table as required, and every sum of these three facts is *valid* and *correct*.

Measured facts in the retail sales schema

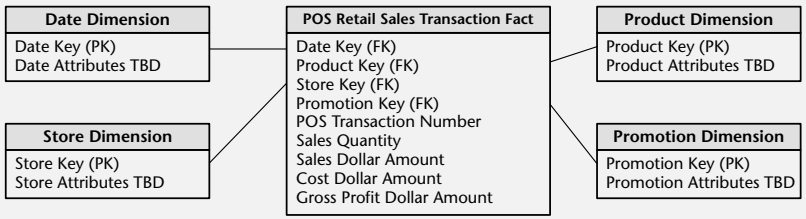


Figure 3: Measured facts in the retail sales schema

- What do we have at this point?
- We have the facts (what we'll measure) fully specified.

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Sample Star Schema

In general, pick 1 dimensional attribute from 1 or more dimensions.

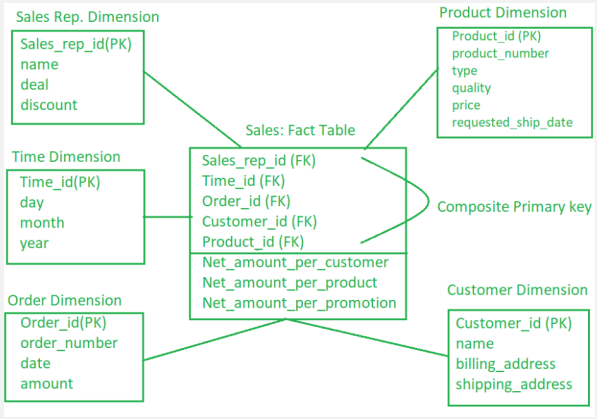


Figure 4: Can you think of 1-D, 2-D, 3-D and 4-D analyses?

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